

UNIVERSITY OF INFORMATION TECHNOLOGY & SCIENCES

Software Requirements Specification (SRS)

IoT-Based Smart Launch Overcrowding Detection & Risk Assessment System

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Submitted From:

Student ID	Name
0432320005101163	Tahsin Sarkar Mome
0432320005101191	Sheikh Tanziya Khanam Sumaiya
0432320005101186	Soniya Chowdhury
0432320005101155	Fahima Faruk Neha

Submitted To:

Engineer Md. Safaet Hossain

Associate Professor & Director, Department of CSE, UITS

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Project: IoT-Based Smart Launch Overcrowding Detection & Risk Assessment System

Table of Contents

1.Introduction

2.Overall Description

3.System Features

4.External Interface Requirements

5. Functional Requirements

6.Non-Functional Requirements

7. Domain Requirements

8.System Architecture

9.ER Diagram

10.Assumptions and Dependencies

11.Future Enhancements

12.Conclusion

1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification for the AIoT-Based Smart Launch Overcrowding Detection & Risk Assessment System.

The system is designed as a real-time IoT and AI/ML-based early warning system for safer river transportation in Bangladesh. It

monitors passenger count, vessel load, tilt, stability, and water level and generates a Safety Risk Score from the collected sensor data.

The system aims to detect dangerous conditions before they become critical and provide timely warnings to the captain, terminal authority, and relevant authorities.

1.2 Scope

The system will monitor passenger overcrowding and vessel stability using IoT sensors installed on passenger launches or ferries.

The system will:

- Count passengers in real time.
- Monitor total vessel load.
- Detect uneven load distribution.
- Monitor vessel roll and pitch.
- Detect water ingress into the hull.
- Track vessel location using GPS.
- Combine sensor information using AI/ML.
- Generate a Safety Risk Score.
- Display Safe, Warning, High Risk, or Critical status.
- Provide multi-level alerts.
- Store sensor data for historical analysis.
- Continue logging data during low/no-network conditions.
- Provide a captain dashboard and remote control-room dashboard.

1.3 Definitions

AIoT: Integration of Artificial Intelligence and Internet of Things technologies.

Overcrowding: Passenger count exceeding the safe/rated capacity of a vessel.

Load Imbalance: Uneven distribution of passengers or weight across different points of the vessel.

Risk Score: AI/ML-generated safety assessment based on sensor data.

Tilt: Angular movement of the vessel caused by imbalance or instability.

Captain: Person responsible for operating the launch.

Terminal Authority: Authority responsible for monitoring launch operations.

Control Room: Central monitoring facility used by relevant authorities.

2. Overall Description

2.1 Product Perspective

The proposed system is an AIoT-based safety monitoring platform consisting of onboard sensors, ESP32 firmware, communication modules, backend services, an AI/ML service, database, and web dashboards.

The system collects sensor data from the launch and transfers it through the communication layer to the backend. The AI/ML service processes the data and generates a Safety Risk Score.

2.2 Product Functions

The major functions of the system are:

- Real-time passenger counting.
- Vessel load monitoring.
- Multi-point load distribution monitoring.
- Tilt and stability monitoring.
- Water-level/bilge sensing.
- AI/ML-based risk scoring.
- Explainable risk output.
- Multi-level safety alerts.
- GPS and route tracking.
- Captain dashboard.
- Remote web dashboard.
- Historical analytics.
- Offline-first data logging.

2.3 User Classes

Captain

The captain can monitor the current safety condition of the launch through the onboard dashboard and receive warnings when dangerous conditions are detected.

Terminal Authority

The terminal authority can monitor launch conditions and receive alerts regarding critical situations.

Control-Room Authority

The authority can monitor multiple vessels remotely through the web dashboard.

System Administrator

The administrator can manage the system, sensor configurations, users, and stored monitoring data.

2.4 Operating Environment

Hardware:

- ESP32
- IR/Ultrasonic sensors
- Load cells
- HX711
- MPU6050 accelerometer/gyroscope
- Water-level sensor
- GPS module
- GSM/LoRa communication modules

Frontend: React.js

Backend: Node.js with Express

ML Service: Python, Flask, Scikit-learn

Database: MySQL

Firmware: C++ using Arduino IDE

2.5 Constraints

- Sensor accuracy may affect the quality of risk assessment.
- Communication networks may be unavailable in remote river areas.

- Hardware installation must be suitable for the launch environment.
- The system depends on appropriate sensor calibration.
- AI/ML predictions depend on the quality of available sensor data.
- Deployment cost must remain practical for individual vessels.

3. System Features

Feature 1: Passenger Counting

The system will use IR/ultrasonic sensors at entry points to count passengers in real time and compare the count with the rated vessel capacity.

Feature 2: Multi-Point Load Monitoring

Load cells installed at multiple points will measure vessel weight and identify uneven load distribution.

Feature 3: Tilt & Stability Monitoring

An MPU6050 accelerometer/gyroscope will monitor vessel roll and pitch to detect dangerous instability.

Feature 4: Water-Level Sensing

The system will detect water ingress into the hull as an early indication of structural danger.

Feature 5: AI/ML Safety Risk Score

The system will combine passenger count, load, tilt, water level, and other available data to generate a safety risk score.

The risk status will be categorized as:

- Safe
- Warning
- High Risk
- Critical

Feature 6: Multi-Level Alerts

The system will provide:

- On-board siren.
- LED warning light.
- Voice announcement.
- Alerts to terminal authority.
- Critical alerts to relevant authorities.

Feature 7: GPS & Route Tracking

The system will track the vessel's location and route.

Feature 8: Captain Dashboard

The dashboard will display:

- Passenger count.
- Load percentage.
- Tilt angle.
- Water level.
- Current risk status.
- Recommended action.

Feature 9: Offline-First Logging

The system will store data locally when network connectivity is unavailable and synchronize the data after reconnection.

Feature 10: Remote Web Dashboard

Authorities can monitor vessel location and safety status through a central control-room dashboard.

Feature 11: Historical Analytics

The system will maintain historical information about load, tilt, and risk to support long-term safety analysis.

4. External Interface Requirements

4.1 User Interface

The system will provide web-based graphical dashboards.

The main interfaces will include:

- Captain Dashboard
- Remote Control-Room Dashboard
- Risk Status Display
- Alert Interface
- Historical Analytics Interface
- Vessel Monitoring Interface

4.2 Hardware Interface

The system will interface with:

- ESP32 microcontroller.
- IR/Ultrasonic passenger-counting sensors.
- Load cells.
- HX711 load-cell amplifier.
- MPU6050 accelerometer/gyroscope.
- Water-level sensor.

- GPS module.
- GSM/LoRa modules.
- Siren.
- LED warning light.
- Voice announcement system.

4.3 Software Interface

The system will use:

- React.js for frontend dashboards.
- Node.js/Express for backend services.
- MySQL for structured data storage.
- Python/Flask for ML services.
- Scikit-learn for machine-learning models.
- C++/Arduino IDE for ESP32 firmware.

4.4 Communication Interface

Communication between the onboard system and server will use GSM and/or LoRa modules depending on network availability.

The system will support offline data logging when network connectivity is unavailable and automatically synchronize stored information after reconnection.

5. Functional Requirements

Functional requirements describe the services and operations that the proposed system shall provide.

FR1: Passenger Counting

The system shall count passengers entering and leaving the vessel using IR/ultrasonic sensors.

FR2: Capacity Monitoring

The system shall compare the current passenger count with the rated vessel capacity.

FR3: Load Monitoring

The system shall collect weight information from multiple load cells installed at different vessel points.

FR4: Load Imbalance Detection

The system shall analyze multi-point load data to identify uneven distribution of weight.

FR5: Tilt Monitoring

The system shall continuously measure vessel roll and pitch using an accelerometer/gyroscope.

FR6: Water-Level Monitoring

The system shall monitor water ingress into the vessel hull and generate a warning when abnormal water levels are detected.

FR7: Sensor Data Collection

The ESP32 shall collect sensor readings and transmit the data to the backend system.

FR8: AI/ML Risk Assessment

The system shall process sensor data through the ML service and generate a Safety Risk Score.

FR9: Risk Classification

The system shall classify the vessel's safety condition into:

- Safe
- Warning
- High Risk
- Critical

FR10: Explainable Risk Output

The system shall identify the major factors contributing to the risk score, such as excessive load, dangerous tilt, or abnormal water level.

FR11: Automatic Alerts

The system shall automatically generate alerts when predefined safety conditions are reached.

FR12: Multi-Level Alert Escalation

The system shall escalate critical alerts from the onboard system to the terminal authority and relevant authorities.

FR13: GPS Tracking

The system shall collect and display the vessel's current location.

FR14: Captain Dashboard

The system shall display real-time vessel safety information to the captain.

FR15: Remote Monitoring

The system shall allow authorized control-room personnel to monitor vessel status remotely.

6. Non-Functional Requirements

6.1 Performance Requirements

- The system should process sensor readings continuously.
- The dashboard should display updated monitoring information with minimal delay.
- Critical safety alerts should be generated as quickly as possible after dangerous conditions are detected.
- The system should support monitoring of multiple vessels through the remote dashboard.

6.2 Security Requirements

- Only authorized users shall access the monitoring dashboards.
- User authentication shall be implemented.
- Role-based access shall restrict users according to their responsibilities.
- Sensor and monitoring data shall be protected from unauthorized modification.

6.3 Usability Requirements

- The dashboard should be simple and easy to understand.
- Risk levels should be clearly displayed.
- Critical warnings should be visually and audibly distinguishable.
- The captain should be able to understand the recommended action without requiring extensive technical knowledge.

6.4 Reliability & Availability Requirements

- The system should continuously monitor vessel conditions during operation.
- The system should continue collecting and storing data during temporary network failure.
- Locally stored data should be synchronized after reconnection.
- Sensor failure or communication failure should not silently stop the monitoring system.

7. Domain Requirements

DR1: Vessel Capacity Safety

The system shall monitor passenger count against the rated capacity of the vessel to help identify overcrowding.

DR2: Stability Monitoring

The system shall monitor vessel tilt and load distribution because uneven loading can contribute to dangerous instability.

DR3: Early Warning

The system shall provide warnings before unsafe conditions become critical.

DR4: River Transportation Environment

The system should support operation in river areas where network connectivity may be weak or unavailable.

DR5: Authority Monitoring

The system shall provide relevant authorities with centralized information about vessel safety conditions.

8. System Architecture

The proposed system follows a multi-layer architecture:

1. Hardware/Firmware Layer

ESP32 collects data from passenger-counting sensors, load cells, MPU6050, and water-level sensors.

2. Communication Layer

GSM and LoRa modules transmit sensor information.

3. Backend Layer

Node.js/Express receives sensor data, handles APIs, and manages alerts.

4. ML Service Layer

Python/Flask and Scikit-learn process sensor data and generate the Safety Risk Score.

5. Database Layer

MySQL stores sensor readings, timestamps, vessel information, and risk records.

6. Presentation Layer

React.js provides captain and control-room dashboards.

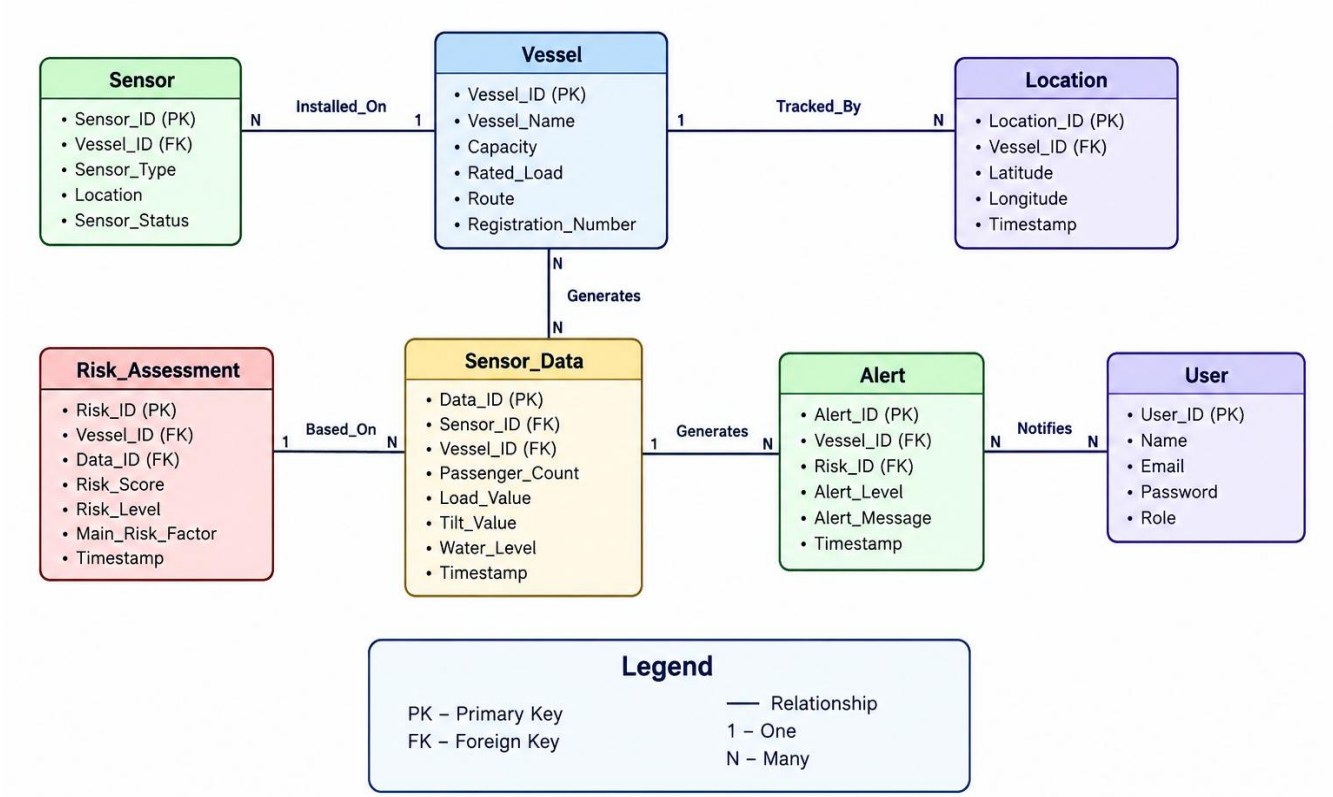
The presentation specifies the overall flow as:

ESP32 → Node.js API → MySQL

and

Node.js → Python ML Service → Risk Score → React Dashboard.

9. ER Diagram



10. Assumptions and Dependencies

The system assumes that:

- Sensors are properly installed and calibrated.
- The vessel provides an appropriate environment for sensor installation.
- The rated capacity of each vessel is available.
- Users provide valid authentication information.
- GPS information is available when location tracking is required.
- GSM/LoRa communication may be available depending on the river area.

- The AI/ML service receives sufficient sensor data for risk assessment.
- Relevant authorities are available to receive and respond to critical alerts.

The system depends on the proper functioning of the ESP32, sensors, communication modules, backend server, database, and ML service.

11. Future Enhancements

Future improvements may include:

- Advanced AI/ML models for improved risk prediction.
- Mobile application for authorities and captains.
- Improved real-time route visualization.
- Integration with additional weather and environmental sensors.
- Automated emergency communication.
- More advanced predictive analytics.
- Integration with centralized national river-transport monitoring systems.
- Improved sensor-fault detection.

12. Conclusion

The AIoT-Based Smart Launch Overcrowding Detection & Risk Assessment System is designed to provide a real-time early warning mechanism for safer river transportation in Bangladesh.

By combining passenger counting, multi-point load monitoring, tilt and stability sensing, water-level monitoring, GPS tracking, and

AI/ML-based risk assessment, the system can identify potentially dangerous conditions at an early stage.

The proposed system also provides multi-level alerts, captain monitoring, remote authority monitoring, offline-first data logging, and historical analytics. Therefore, the system aims to support safer vessel operation and provide a data-driven approach to reducing risks associated with overcrowding, load imbalance, and vessel instability.